1 Coordinates

Definition 1.1 (Ordered basis)**.** Let V be a finite-dimensional vector space. An *ordered basis* of V is a basis $B = \{v_1, v_2, \ldots, v_n\}$ together with a chosen ordering of the basis vectors. We denote an ordered basis as an n-tuple $(v_1, v_2, \ldots, v_n)$.

Let V be a finite-dimensional vector space over F and $B = (v_1, v_2, \ldots, v_n)$ an ordered basis of V. We proved last time that the linear combination map $L_B : F^n \to V$ defined by

$$L_B(x_1, x_2, \ldots, x_n) = x_1 v_1 + x_2 v_2 + \cdots + x_n v_n$$

is an isomorphism. We will call this the *basis isomorphism* with respect to B. The inverse isomorphism $L_B^{-1} : V \to F^n$ takes a vector $v \in V$ to the point $(x_1, x_2, \ldots, x_n) \in F^n$, which are the coefficients of v with respect to the basis B. We call L_B^{-1} the *coordinate isomorphism* with respect to B. We call $[v]_B := (x_1, x_2, \ldots, x_n) \in F^n$ the *coordinate vector of v with respect to B*. When B is understood, we write $[v]$ for the coordinate vector of V.

Remark 1.2. Note that the use of an ordered basis here allows us to specify a first coordinate, second coordinate, etc. of $[v]$.

Example 1.3. Let $V = F^n$ and $B = (e_1, e_2, \ldots, e_n)$, where e_i is the n-tuple with a 1 in the ith position and 0s elsewhere. Then for $x = (x_1, x_2, \ldots, x_n) \in V$,

$$L_B(x) = L_B(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n = (x_1, \ldots, x_n) = x$$

which shows that $L_B = \mathrm{id}_{F^n}$ (in which case we also have $L_B^{-1} = \mathrm{id}_{F^n}$).

Example 1.4. Let $V = \mathscr{P}_n(F)$ and $B = (1, x, x^2, \ldots, x^n)$. Then for $p = a_0 + a_1 x + \cdots + a_n x^n \in \mathscr{P}_n(F)$,

$$[p]_B = L_B^{-1}(a_0 + a_1 x + \cdots + a_n x^n) = (a_0, a_1, a_2, \ldots, a_n) \in F^{n+1}.$$

Remark 1.5. Since $F^n \cong M^{n\times 1}(F)$, we often write $(x_1, \ldots, x_n)$ interchangeably with $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ without comment.

2 Matrix of a Linear Transformation

We now show that every linear map $T : V \to W$ between finite-dimensional vector spaces can be represented by a matrix. Choose ordered bases $B_1 = (v_1, \ldots, v_n)$ and $B_2 = (w_1, \ldots, w_m)$ of V and W, respectively. Since $Tv_j \in W$, there exist unique scalars T_{ij} such that

$$Tv_j = \sum_{i=1}^{m} T_{ij} w_i. \tag{2.1}$$

Definition 2.2 (Matrix of a Linear Map)**.** We define the *matrix of T with respect to bases B_1 and B_2*, denoted by $[T]_{B_2,B_1}$, to be the $m \times n$ matrix with entries T_{ij} defined by (2.1). From (2.1), we see that for $j = 1, \ldots, n$, the jth column is $[Tv_j]_{B_2}$, the coordinate vector of Tv_j with respect to B_2. When B_1 and B_2 are understood, we write $[T]$ for the matrix of T.

Example 2.3. Let $V = \mathscr{P}_1(F)$, $W = \mathscr{P}_3(F)$, and $T : \mathscr{P}_1(F) \to \mathscr{P}_3(F)$ be defined by $(Tp)(x) = x^2 p(x)$. Choose $B_1 = (v_1, v_2) = (1, x)$ and $B_2 = (w_1, w_2, w_3, w_4) = (1, x, x^2, x^3)$ to be the standard ordered bases of V and W. We have

$$Tv_1 = x^2 \cdot 1 = x^2 = w_3,$$
$$Tv_2 = x^2 \cdot x = x^3 = w_4,$$

and therefore

$$[Tv_1]_{B_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad [Tv_2]_{B_2} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since $[Tv_i]_{B_2}$ is the ith column of the matrix of T,

$$[T]_{B_2,B_1} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

We will now show that applying a linear map T to a vector v is the same as multiplying the coordinate vector of v by the matrix of T. If $v \in V$ then we can write $v = \sum_{j=1}^{n} ([v])_j v_j$. Then

$$\begin{aligned} Tv &= T\Big(\sum_{j=1}^{n} ([v])_j v_j\Big) \\ &= \sum_{j=1}^{n} ([v])_j Tv_j \quad \text{(since } T \text{ is linear)} \\ &= \sum_{j=1}^{n} ([v])_j \sum_{i=1}^{m} T_{ij} w_i \quad \Big(\text{since } Tv_j = \sum_{i=1}^{m} T_{ij} w_i\Big) \\ &= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} T_{ij} ([v])_j \right) w_i \quad \text{(commutative and associative properties in } F), \end{aligned}$$

or

$$([Tv])_i = \sum_{j=1}^{n} T_{ij} ([v])_j. \tag{2.4}$$

In other words, (2.4) says that the coordinate vector of Tv is given by multiplying the coordinate vector of v by the matrix of T: $[Tv] = [T][v]$.

Example 2.5. Consider again $T \in \mathscr{L}(\mathscr{P}_1(x), \mathscr{P}_3(x))$ defined by $(Tp)(x) = x^2 p(x)$. If $p(x) = a_0 + a_1 x$, then $(Tp)(x) = a_0 x^2 + a_1 x^3$. In Example 2.3 we found the matrix of T is given by

$$[T] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The coordinate vector of $p(x) = a_0 + a_1 x$ is $[p] = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$. Multiplying $[T][p]$ gives

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ a_0 \\ a_1 \end{bmatrix},$$

which is the coordinate vector of $x^2 p(x) = 0 + 0x + a_0 x^2 + a_1 x^3$.

Exercise 2.6. The matrix of a linear map $T \in \mathscr{L}(F^n, F^m)$ with respect to the standard bases of F^n and F^m is called the *standard matrix* of T. Find the standard matrix of $T \in \mathscr{L}(F^2, F^3)$, where T is defined by

$$T(x,y) = (x+3y, 2x+5y, 7x+9y).$$

Exercise 2.7. Let $D \in \mathscr{L}(\mathscr{P}_3(\mathbb{R}), \mathscr{P}_2(\mathbb{R}))$ be the differentiation map defined by $Dp = p'$. Find the standard matrix of D check that $[p'] = [D][p]$ when $p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$.

3 Dimension of $\mathscr{L}(V,W)$

We proved that $\mathscr{L}(V,W)$ is a vector space with respect to the operations

$$(T_1+T_2)(v) := T_1v + T_2v$$
$$(\alpha T)(v) := \alpha Tv$$

We will now prove that when V and W are finite-dimensional with $\dim V = n$ and $\dim W = m$, $\mathscr{L}(V,W)$ is finite-dimensional with $\dim \mathscr{L}(V,W) = mn$.

Exercise 3.1. Fix ordered bases for V and W and consider the map $\psi : \mathscr{L}(V,W) \to M^{m\times n}(F)$ defined by $\psi(T) \mapsto [T]$. That is, ψ takes a linear map T to its matrix $[T]$ with respect to these two ordered bases. We will prove that ψ is an isomorphism. Since two vector spaces are isomorphic if and only if they have the same dimension, it follows that $\dim \mathscr{L}(V,W) = \dim M^{m\times n}(F) = mn$.

(a) Prove that ψ is linear.

(b) Prove that ψ is injective.

(c) Prove that ψ is also surjective, and hence an isomorphism.

4 Product of Linear Maps

We previously defined a product of linear operators in $\mathscr{L}(V)$ by composition: $T_2T_1 = T_2 \circ T_1$.

Exercise 4.1. Assume V is finite-dimensional and let $(v_1, \ldots, v_n)$ be an ordered basis of V.

(a) Show that $[\mathrm{id}_V] = I_n$, the $n \times n$ identity matrix.

(b) Show that $[T_2T_1] = [T_2][T_1]$. That is, the matrix representing the linear operator T_2T_1 is the product of the matrices representing the linear operators T_2 and T_1.

In other words, the vector space isomorphism $\mathscr{L}(V) \to M^{n\times n}(F)$ from the previous section is actually an isomorphism of unital associative F-algebras.

Exercise 4.2. Let $V = \mathscr{P}_2(\mathbb{R})$ with the standard ordered basis. Consider $T \in \mathscr{L}(V)$ defined by $(Tp)(x) = xp'(x)$. Verify that $[T^2] = [T]^2$.